



Avradip Mazumdar Software Engineer | M.Eng. Information Technology

✉ rontymazumdar@gmail.com ☎ +49-15510440410 📍 Wiesbaden 🔗 LinkedIn 🐙 GitHub

Software engineer with 2.5+ years of industry experience at Cognizant Technology Solutions building and maintaining backend services and cloud-deployed applications. Now pursuing an M.Eng. in Information Technology at Frankfurt UAS and actively seeking a Working Student position to further deepen my expertise in technologies.

🧠 SKILLS

PROGRAMMING: Python, SQL, Java, C, C#

BACKEND & APIs: REST API, FastAPI, Flask, Django, Numpy, Pandas, Matplotlib, BeautifulSoup, async Python, HTTP fundamentals, .NET / ASP.NET

FRONTEND & WEB: React.js, JavaScript, TypeScript, HTML5, CSS3

DATABASES: PostgreSQL, MySQL, MS-SQL, PL/SQL, SQLAlchemy ORM

TECHNOLOGY: Databricks, Azure Data Factory, Azure Container Instances, AWS Lambda, Docker, Kubernetes, GitLab CI/CD, Jenkins, Airflow, Jira, Git/Github, Linux (Ubuntu)

🌐 LANGUAGES

ENGLISH: Fluent (C1)

GERMAN: Conversational (B2)

🎓 EDUCATION

FRANKFURT UNIVERSITY OF APPLIED SCIENCES, M.ENG IN INFORMATION TECHNOLOGY
10/2024 – Present
CGPA -2.5/4

MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, B.TECH IN ELECTRONICS AND COMMUNICATION ENGINEERING
08/2018 – 06/2022
CGPA: 8.84/10

📜 CERTIFICATES

Complete Python 3 Materclass Journey: Udemy

REST API with Python and Django: Udemy

Python Automation Testing with Pytest: Udemy

🏆 AWARDS

MASTERING SKILLS, COGNIZANT TECHNOLOGY SOLUTIONS
03/2024

Certified on Python specialized package

💼 WORK EXPERIENCE

BACKEND & DATA ENGINEER, COGNIZANT TECHNOLOGY SOLUTIONS

06/2022 – 09/2024 | Kolkata, India

- **Backend service development:** Engineered scalable Python backend services and FAST APIs with full unit-test coverage, 100% docstring documentation, and 99.9% uptime across enterprise deployments.
- **Database & SQL work:** Implemented complex SQL procedures and ORM queries across PostgreSQL and MySQL, including joins, aggregations, and query optimisation over multi-million-record tables for analytical and reporting consumers.
- **Process improvement:** Proposed and shipped automation that removed manual steps from recurring data-delivery workflows, with the same review-and-iterate mindset that an internal-consulting role requires.
- **Cloud deployment:** Led migration of legacy services to cloud-native Azure architectures via reproducible GitLab CI/CD workflows.

Tech Stack: Python, FastAPI, Numpy, Pandas, SQLAlchemy, Pydantic, pytest, Azure Data Factory, Databricks, PostgreSQL, MySQL, GitLab CI/ CD

PROGRAMMER ANALYST INTERN, COGNIZANT TECHNOLOGY SOLUTIONS

02/2022 – 06/2022 | Kolkata, India

- Contributed to large-scale distributed data processing with Apache Spark; documented modules thoroughly and participated in structured code reviews.

Tech Stack: Java, Apache Spark, SQL, SparkSQL

📁 PROJECTS

CNN-BASED REAL-TIME OBJECT DETECTION 🌐,

Tech Stack: Python, PyTorch, TensorFlow, CNN, NumPy, pandas

10/2025 – 03/2026

- Built the full analytical pipeline feeding a machine-learning model — raw data acquisition, cleaning, labelling, feature engineering, and train/test splitting — in Jupyter notebooks with reproducible experiment tracking.
- Trained and evaluated a CNN classifier for 5-class detection, benchmarked with precision, recall, F1, AUC, and mAP; iteratively tuned hyperparameters to reduce false positives.

INTELLIGENT RECRUITMENT WORKFLOW PLATFORM 🌐,

Tech Stack: Java, React.js, Camunda BPMN, GitLab, Agile

10/2025 – 02/2026

- Architected end-to-end backend workflow orchestration in Java with Camunda BPMN, modelling a real domain (recruitment workflow automation) with clear process definitions, decision points, and state transitions.
- Developed a multi-factor matching algorithm in Java (skills, wage, geospatial features) implementing weighted scoring logic against a structured domain model.
- Integrated the Java backend with a React.js dashboard for status tracking; managed version control, branches, pull requests, and reviews via GitLab.

INVESTIGATE INPUT RECONSTRUCTION USING CLASSIFIERS 🌐,

Tech Stack: C#, .NET 9, Docker, Azure Container Instances

04/2025 – 08/2025

- Built a backend service exposing an image-classification API in C# on .NET 9, including the service logic, integration of HTM Spatial Pooler and KNN classifier components, and structured logging.
- Containerised the .NET service with Docker and deployed to Azure Container Instances, managing the full deployment lifecycle including monitoring and iterating on runtime stability.
- Implemented automated validation checks and reproducibility safeguards across experiments.